GROTHENDIECK TOPOSES: ARCHITECTURAL AND PLASTIC IMAGINATION BEYOND MATERIAL NUMBER AND SPACE

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Abstract. The first part of the article introduces the work of Alexander Grothendieck (1928-2014) from a conceptual standpoint, focusing on his *topos theory* (1962), a general setting which encompasses both arithmetic (*number*) and geometry (*space*) under a common abstract perspective (*sheaves*: a far-reaching tool which helps gluing the local and the global). Grothendieck's revolution, wider than Einstein's interlacing of space and time, has radically changed mathematics, but its *plastic* influence beyond the specialty has yet to be developed. Focusing on that program, the second part of the article explores how Grothendieck toposes can be used to understand the *imaginative* clash between the visible and the invisible, between materiality and non-materiality in the works of Andrei Tarkovsky (cinema), Frank Gehry (architecture) and Anselm Kiefer (art).

Keywords. Mathematics, topos theory, architecture, art, film studies.

1. A QUICK GLIMPSE ON GROTHENDIECK'S WORK

Alexander (=Alexandre) Grothendieck (Berlin 1928 – Saint-Girons 2014) must be considered, without doubt, as the major mathematical genius of the last sixty years, and, next to David Hilbert, as one of the two fundamental mathematicians of the last century. With a published work of near *ten thousand* pages (covering geometry, topology, number theory, complex variables, to only quote the *heart* of mathematics), with more than *one thousand* new definitions (when an ordinary mathematician may be happy if she has introduced *one* new definition in the field), with complete revolutionary understandings of the notions on *number* ("schemes"), *space* ("topos") and *form* ("motives"), Grothendieck has opened all type of new roads for the development of mathematics in the following centuries.

Nevertheless, Grothendieck's work is almost completely unknown outside a small group of specialists. In contraposition with Einstein, whose ideas (simplified or deformed) have attained the public domain, Grothendieck still lies in the shadow, even if his conceptual revolution goes much farther. In fact, if Einstein studies the

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relativization of *space-time* and discovers its local invariants, Grothendieck studies the relativization of *space-number* and discovers its *global universal* invariants. Around his recent death, a sudden interest has arisen for a life worth of fiction (partial biographies [Bringuier 2015], [Pradeau 2016], [Douroux 2016], and an inspiring novel [Fonseca 2015]), but time is still needed to fully appreciate the magnitude of his mathematical work.

Grothendieck's inexhaustible universe may be divided (wrongly, as in any compartmentalization) in *four main periods*. *(I).* From 1949 to 1957, along certain geographical *margins* (Nancy, Sao Paulo, Kansas), he produces fundamental contributions in topological vector spaces, homology, category theory, complex variables, obtaining profound theorems [Grothendieck 1955, 1956, 1957a, 1957b], and leaving rich seeds to be developed in the following decades. *(II).* Between 1958 and 1970, in the very *center* of mathematics (Paris), Grothendieck advances his research at the IHES (*Institut des Hautes Études Scientifiques*), specially constructed to shelter him. After a scintillating, condensed, vision of his future programs [Grothendieck 1958], he writes with the formidable Dieudonné the *Éléments de Géométrie Algébrique* (EGA) (appearance of *schemes*) [Grothendieck 1960-66], and he directs his famous *Séminaire de Géométrie Algébrique* (SGA) (appearance of *toposes*) [Grothendieck 1960-69]. An archetypical vision of form, a common root for all (co)homologies (appearance of *motives*) [Grothendieck c. 1967], emerges also in the IHES years. The finer mathematical grain of the epoch goes through the Institute, where innovation becomes a trademark: a nice anecdote tells that, as a storm-head visitor complained about the meager IHES library, Grothendieck replied – "here we do not read mathematics, here we make them".

(III). After leaving the IHES in 1970 (political radicalization, intemperance, disappointment with his entourage), Grothendieck founds a radical ecologist movement *Survivre et vivre* ("survival" first, "survival and life" afterwards), to which he offers the best of his indomitable energy. He goes away from the mathematical community, returns to the *margins* (Université de Montpellier, where he had done his undergraduate studies, 1945-1948), and hides happily in the French province. Between 1981 and 1991, a passion for mathematics surges again, along extraordinary manuscripts [Grothendieck 1981, 1983, 1984, 1991] (moduli spaces of Riemann surfaces, fundamental groupoids, anabelian geometry, tame topology, *dessins d'enfants*, derivators, etc.) On another hand, he writes an intense (and extensive) mathematical reflection *Récoltes et semailles*

[Grothendieck 1986], where he attacks forcefully a mathematical community that (in his view) had betrayed him, but, above all, where he elaborates the major treatise ever conceived about *mathematical creativity*. *(IV).* In 1991 Grothendieck disappears, and secludes completely himself in the Pyrenees (the last *margin*). Nevertheless, between 1991 and 2014, he continues steadily to write, and finally leaves *fifty thousand* manuscript pages to the *Bibliothèque Nationale de France* (a legacy now disputed by his family). A report by Georges Maltsiniotis (March 2016) describes the existence of ten thousand mathematical pages, and some thirty thousand pages on a *Treatise on Evil*. The sheer monumentality of Grothendieck's enterprises surpasses our usual canons, and one or two decades will be needed to have a reasonable account of this last legacy.

The interest of modern (1830-1950) and contemporary (1950-today) mathematics, where Grothendieck lays, consists, not so in the partial modelings that they may offer, but rather in its *help to deploy imagination*. A hundred years before Grothendieck, Riemann surfaces had stimulated visual and conceptual inventiveness [Riemann 1851]. Full of *plasticity*, the very *handling* of Riemann surfaces (they are beautiful *material* constructs), allows transits and possibilities, that rigid, classical, non-plastic geometries prevent. On another hand, Riemann had observed the importance of *analytic continuity* for the development of functions of a complex variable (holomorphic and meromorphic functions). A hundred years later, *sheaves* were invented by Leray (at a concentration camp, 1942, at the same time that the young Grothendieck was interned with his mother in another camp), and were further developed in Cartan's seminar at the *École normale supérieure* (where Grothendieck landed after his undergraduate career). A sheaf is the *simplest* mathematical object (just two topological spaces related by a projection with a good local behavior), which can be oriented towards a precise study of *local/global* processes. In the language of sheaves, Riemann's analytic continuity is expressed by the structural fact that the sheaf of germs of holomorphic functions is *separated*. Thus, a sophisticated dialectics –elasticity/rigidity, continuity/separation– enters into play (see *Figure 1*).

In a similar vein, *Grothendieck toposes* (categories equivalent to categories of sheaves over abstract topologies, 1962) constitute *plastic sites*, specifically open to dynamic variations. Grothendieck toposes unify deep insights on arithmetic (number) and geometry (space) [Grothendieck 1960-1969]. Beyond Cantorian, classical, static sets, the objects in a topos are to be understood as generalizations of *variable sets*. Instead of living over a rigid bottom, governed by classical logic, they live over a dynamic Kripke model, governed by *intuitionistic* logic. Beyond the classical example of the separated sheaf of holomorphic functions, a sheaf does *not* have to be separated in a general topos: *points do not have to determine their associated objects*. We can even imagine objects without points, defined only through *flux processes*. A wonderful example is the topos of *actions* of monoids. Such a topos has an underlying classical logic (where the law of excluded middle holds and points are essential) if and only if the monoid is a group. Thus, when we deal with structures which are *monoid non-groups*, the logic of their action is just *intuitionistic*, *non-separated*, closer to topological fluxions, deformations, disruptions.

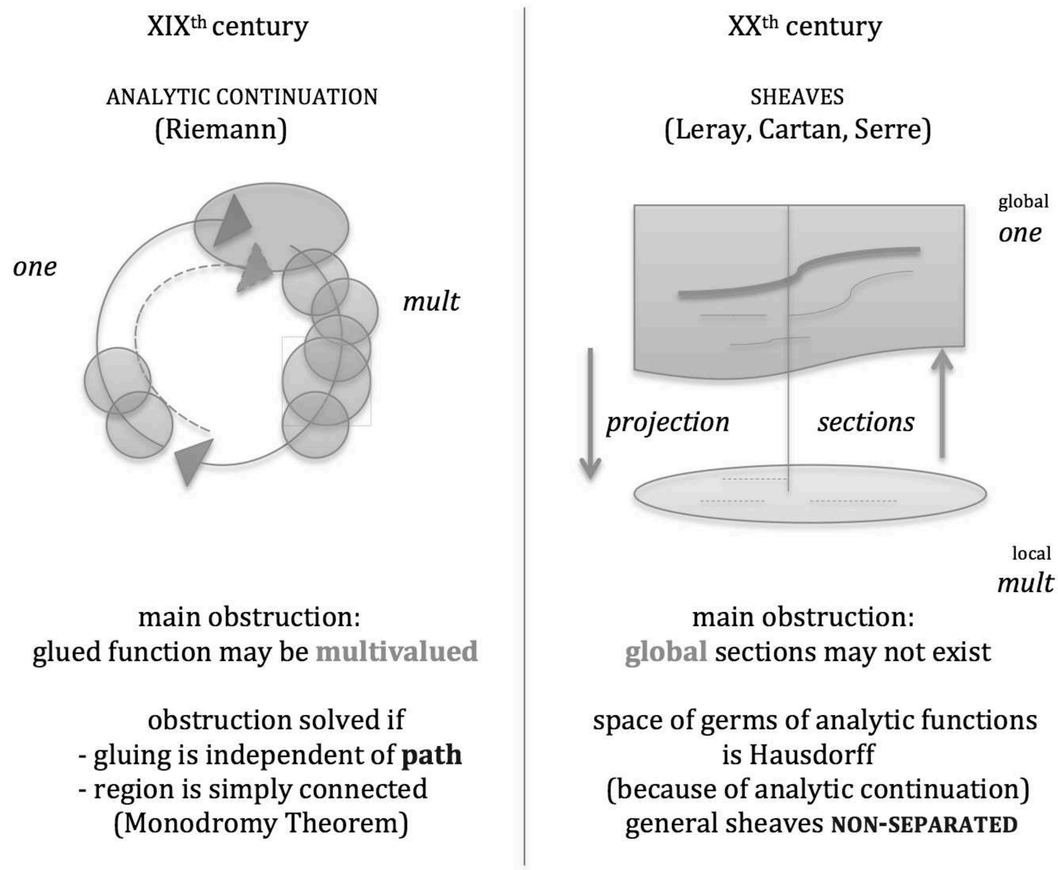

Figure 1
The sheaf concept

In a simple *motto*, we may characterize Grothendieck as the greatest mathematician who, after Poincaré, unified Galois and Riemann, the two mathematical geniuses of XIXth century (see [Zalamea 2009] for an extended study of Grothendieck's work). We may thus imagine that, on a conceptual, non exclusively technical level, Grothendieck toposes generalize in some sort Riemann surfaces. Even if the assertion is not mathematically correct (we would have to introduce also Grothendieck *schemes* into the account), *methodologically* it is fruitful. *Figure 1* shows dynamic, relative movements. *Bases change*, no eternal objects are considered. *But* the correlative changes are in turn studied mathematically, and they are incorporated as new objects in an *upper level*. In this way, emerges an *iterated dynamics* –a dynamical study of dynamics– which can have a strong influence in cultural studies.

As another example, going a little further and *inverting* our thought, one can deal with toposes whose underlying logics are *dual* to intuitionistic logic. *Paraconsistent logics* (1963) thus appear in the panorama, logics where we can have *local* contradictions *without* forcing *global* contradictions which would destroy the system [Costa & Béziau 1997]. It is fairly clear that the dynamic *logics of art* are either intuitionistic or paraconsistent, but certainly not classical. The dialectics of art and mathematics already hint to such a situation, but it is interesting to notice that, (1) if we situate ourselves in a multilayered site akin to art, that is, in some sort of non-well defined Grothendieck topos of *art actions* (compare with the well-defined topos of monoid actions), (2) if we think *dually* –as art should do with respect to mathematics–, and (3) if we allow ourselves to live in a continuous *medium* of contradictions, then we will be approaching many of the strongest forms of contemporary art. We will be entering into that perspective in the following section.

2. Grothendieck toposes along cinema, architecture and art

Using *synthetic* and *multiple* perspectives applied to the *sheaf* concept, Grothendieck obtains an extended notion of *place* (*topos*). Two major shifts are indeed proposed: *(i)* consider not an usual, analytic, topology (given by open sets), but an alternative, synthetic, topology (given by *arrow coverings*: sets are replaced by relations); *(ii)* consider not one sheaf, but *all* sheaves of a given kind (a *category* of sheaves) on such a synthetic topology (called a "Grothendieck topology"). A category of sheaves over a

Grothendieck topology is, by definition, a *Grothendieck topos*. Now, many Grothendieck topologies are available on many categories, and a variation of the topologies gives a *variation of the sites* (which, in turn, yield *toposes).* Some Grothendieck topologies produce *algebraic and spatial sites* (such as the *étale* topos of a scheme, with which Grothendieck and his school solved the Weil conjectures), others produce *arithmetical sites* (such as the arithmetical topos of Alain Connes, with which he hopes to structure better and perhaps solve the Riemann Hypothesis, the greatest open problem in mathematics, *cfr.* [Connes 2015]). What interests us here are the *architectonics* of the constructions: a single, *universal*, concept (sheaf) is projected *both* onto space and number, unifying the major, apparently distant, fields of geometry and arithmetic. *Beyond material number and space, Grothendieck toposes render thus visible the invisible.* Before Grothendieck, some visionaries (Galois, Riemann, Poincaré, Florensky, for example) had *imagined* such a common kernel of arithmetic and geometry, but no one had really *seen* it, inside a precise, technical, architectonics. In what follows, we will follow other paths where *the invisible is rendered visible* through other specific architectonics –in cinema (Tarkovsky), architecture (Gehry), and art (Kiefer)– and where some *material gestures* can be related to similar strategies in Grothendieck toposes.

2.1. Tarkovsky: images of the unseen

Andrei Tarkovsky (1932-1986) embeds himself in a very concrete reality, to let us emerge towards the ineffable: "The image is linked to the concrete, to the material, in order to reach afterwards, by mysterious ways, the spirit beyond" [Tarkovsky 1985, p. 136]. The "*image-observation*, the true concrete image" constitutes the support of his entire work: "the image is an impression of the Truth given for us to perceive with our blind eyes" [Tarkovsky 1985, p. 123]. We can feel the *tour de force* proposed by Tarkovsky: there is a non-material domain of Spirit and Truth, that we cannot see with our *blind eyes*, but that we can reach through a mysterious *transmutation* of material images (for a similar, quasi mystical experience in the works of Pavel Florensky, see [Zalamea 2013]). As in the stutterer of *Mirror* (1974), who after an hypnotic session suddenly reaches an *étale* ("flat") language, the "stuttering" of our blind eyes can be overcome when we contemplate the *universe in all its multiplicity*. In that vein, Tarkovsky reveals some of the *Mirror*'s secrets: the "negative charm" of a Leonardo portrait, seen "from outside, or sidewise, with a view coming from above the world";

the possibility to introduce "a part of eternity in the instants deployed under our eyes"; the need to present an artistic image as an "experience of complex, contradictory, emotions"; the desire to walk along that "fleeting instant where the positive ceases to be, where it slips to the negative, and vice versa" [Tarkovsky 1985, pp. 126-127].

The *threshold* becomes thus a central figure in Tarkovsky's architectonics (*cfr.* [Devidts 2012]). Indeed, only *from a threshold* different perspectives become available, and a true multiplication occurs in front of our blind eyes. Reason and sensibility merge together, coming *from above*. In a similar sense, in Grothendieck toposes, number (coming from *discrete* arithmetic) and magnitude (coming from *continuous* geometry) merge together, and may be derived from the abstract sheaf concept. Even more strikingly, in the threshold of Grothendieck's *géométrie arithmétique* (motives and anabelian geometry, going beyond the SGA period on *géométrie algébrique*), higher revelations occur (for other, more mystical revelations, see [Grothendieck 1987]). Tarkovsky's visions –"view coming from above", "part of eternity", "experience of complex", "slip to the negative"– become in Grothendieck's hands deep open conjectures for the development of mathematics (standard conjectures, anabelian conjectures, *cfr.* [Grothendieck c. 1967, 1984]). *Abstraction* (a view from above) becomes an essential *creative force* because, in the general, particular walls disappear. Indeed, *plasticity* reigns in the general (non-material) realm, precisely because we can escape from the constraints of the particular (material) world. And an *inversion of thought*, fundamental for both Grothendieck and Tarkovsky, suggests that an access to non-materiality (a True goal) is obtained through a profound immersion in concrete materiality. Only living on the *threshold of the opposites* (or, in some *True antinomical thinking* would say Florensky, *cfr.* [Zalamea 2013]) may open our blind eyes.

In *Andrei Rublev* (1966), the links of concrete particulars with the "Unique" are manifested all along the film. The first images –ascent and fall– are materialized in a counterpoint of the four elements, when a man on a balloon flies over the river and finishes to crash: fire (balloon), air (flight), water (river) and earth (floor) are fused in a sequence of cinematographic planes which ends in the unified beauty of a horse wallowing in slow motion. On another hand, the last image of the film comes back on the horses, after wandering through many concrete details of Rublev's *Trinity* –lines, mixtures, textures, veils, cracks– through which Tarkovsky opens up our *blind eyes* to somewhat "beyond". All in *Andrei Rublev* expresses the essential contradiction of the threshold between *here and there*: fields, birches, roots, bushes, sands, muds, rains, fumes –preludes to the fantastic material construction of the bell– are signs of a "spirit beyond". It is remarkable that Tarkovsky's "negative metaphysics" is also perfectly reflected in his cinematographic technique. Through his long shots without cuts, his many views without horizon, his montages without defined time fluxions, his washed colour spectra, his contrapositions of silence and sound, the filmmaker always points out to *what is missing*. Looking carefully at the extreme concrete particulars, we can imagine what eludes us. *Stalker* (1979) explores such a Zone of access to an "spirit beyond". All types of *fragments* and *residues* (which recall Benjamin's *Arcades* project, *cfr.* [Benjamin 1927-1940]) are scattered through the Zone, as are the residual dialogues between the Writer (limited by language) and the Professor (limited by science). Straight lines are impossible in the Zone, one always has to walk along broken diagonals, "everything changes every minute". Mixtures, inversions, *vibrations*, are mandatory in the search for a final answer, never reached, never reachable (for extensions of these ideas, see [Zalamea 2010]). The magnificent long shot where we see how the Stalker's daughter advances in front of our eyes, even if we know that she cannot walk, until the camera goes down and we observe that his father is carrying her, evokes all those *thin touches* of miracle and concretion, illusion and despair, joy and pain, life and death, in which we move.

Grothendieck looked himself as an *architect* in the IHES period, the builder of the *abstract* foundations EGA-SGA. Afterwards, he intervened also into the (very modest) *concrete* country houses in which he lived in his retirement periods (Villecun, Mormoiron, Lasserre). Tarkovsky did also construct his house in Myasnoye, in remembrance of his infancy and of the spaces recreated in *Mirror*. A polaroid taken by Tarkovsky (see *Figure 2*, where we have washed away the colours, to emphasize structure) shows the *threshold* of a door, from which a fence, an oak, and a dense fog are visible. It is as if the *greatest visionary architects of contemporary mathematics and cinema* needed, in counterpoint, the simplest, pedestrian, architecture for their daily lives. The tensions between the abstract and the particular, the non-material and the concrete, the invisible and the visible (already deeply explored in [Merleau-Ponty 1964a, 1964b]), acquire here new forms of expression. It is no accident that such a dialectics between work and life may be considered as a *threshold between interior and exterior*. Along that ever-changing *pendulum*, Grothendieck and Tarkovsky lean towards the difficult understanding of an *internal soul*, governed by abstraction, non-materiality, and invisibility.

Figure 2
Myasnoye (September 26 1981) [Tarkovsky 2004, p. 29]

2.2. Gehry: foldings of the unformed

The work of Frank Gehry (b. 1929) is directly inspired in the *freedom* and *plasticity* of the dynamics of motion in art. As Gehry himself reckons, "Panting and sculpture influence my work. For instance, when I had the Bellini picture with the Madonna and Child, I originally thought of it as the Madonna-and-Child strategy for architecture. You see a lot of big buildings with a lot of little buildings, little pavilions in front. (...) I went back to the Bellini and fastened on the folds of the drapery. You see that kind of motion in Giotto, too. A lot of that folding interests me" [Gehry 1999, p. 44]. Gehry goes on to acknowledge the influence of Greek sculpture, Sluter, Brancusi, Moses, or Oldenburg, creative spaces where Gehry finds artistic oscillations, to be transformed in architectural *undulations*. In fact, as Gehry recalls, "I was looking for movement earlier, and found it in the fish. The fish solidified my understanding of how to make architecture move" [Gehry 1999, p. 44]. The fist forms of fishes in Gehry's architecture –Smith residence (1981), lamps (1983-1986), Walker installation (1986), *Fishdance* restaurant (1986)–

evolve towards Barcelona's *Fish Sculpture* (1989-1992), a key moment in Gehry's morphological vision. With the *Fish Sculpture*, architecture is *freed from the weight of materials*. Thanks to a fine use of computer aided design (CATIA, *Computer Aided Three-dimensional Interactive Application*, coming from the French aeronautical industry), a controlled *distribution* of forces and a precisely oriented *assembly* of matter are obtained, helping to construct and elevate curvatures previously impossible to realize.

Leibniz' dream, according to which mathematical symbols would *free imagination*, acquires an entire new dimension with the *Guggenheim Bilbao Museoa* (1991-1997) (for an overview of Gehry's work, see [Gehry 2001]). Heavy matter converted into aerial foldings and unfoldings, thanks to CATIA based mathematical calculations, is one of the most innovative aspects of Gehry's architecture. The *dance* of the titanium ceilings, flying over the "little pavilions in front", allows a contraposition of flow (fish and flower forms) and rigidity (office cubes). *Bilbao*'s atrium encapsulates a forceful dialectics of materiality/non-materiality, combining an *hymn to matter* –water (giant windows over the river and the sky), air (flight of monumental white molds), fire (exterior play of flames and fumaroles), earth (floor and stone walls)– with the outstanding *speed and lightness* liberated through the many perspectives of the site. As Gehry comments, "Everything connected with everything else seems freer, not taking your hands off. I love the free-flow" [Bruggen 1997, p. 37].

Beyond the massive technical realization of *Guggenheim Bilbao*, the *free-flow* emerges in fact with a *liberation of gesture*, to be found in Gehry's initial *sketches* for all his projects [Gehry 2008]. The pendulum between artistic creativity and technical implementation is always present. A *free hand*, liberated even further by CATIA, is one of Gehry's trademarks (see *Figure 3*). Grothendieck's *imaginative freedom* is here well reflected, both in the abstract and the concrete. Through *sketch abstractions*, both the mathematician and the architect envision archetypical non-material entities, which afterwards becomes embodied in *concrete* material types (numbers, spaces, forms). Gehry's drawings and Grothendieck's diagrams allow a *growth* of new ideas, with all sorts of *transits* between the conceptual and the concrete. Gehry's "transient quality", "always exploring forms and how they connect" [Bruggen 1997, pp. 62, 73], resounds thus with Grothendieck's dynamical approach to mathematics, and with his *transient* techniques between different categories.

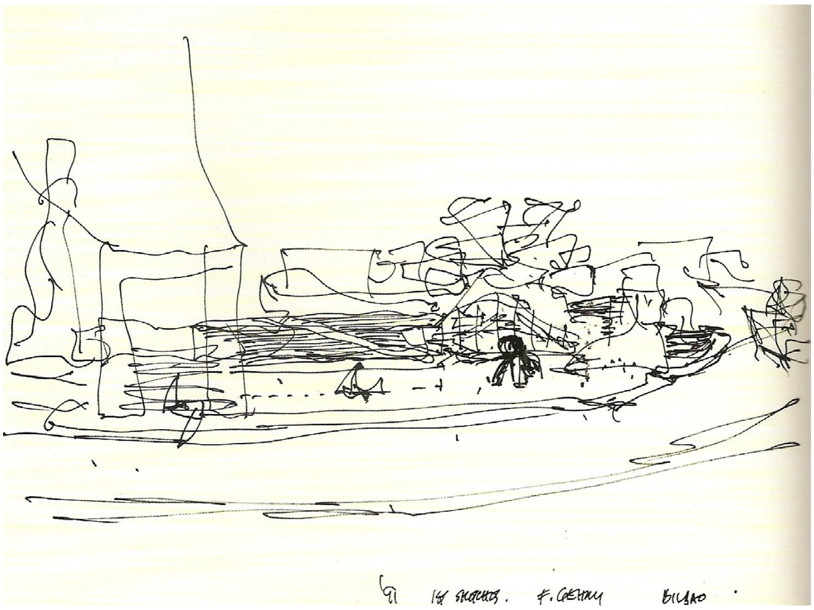

Figure 3
Guggenheim Museum Bilbao (1981) [Gehry 2008, p. 50]

Bilbao's *multiplicity*, through a multitude of different perspectives offered to the spectator, is a perfect material analogue of non-material Grothendieck topologies. The richness of arrows and *coverings* in a Grothendieck topology is embodied in the wide variation of viewpoints and images of the architectural site. Going even further, the *sheaves* over the Grothendieck topology may be seen as those *global* comprehensions of the *Bilbao* structure where all *local* information (walks, senses, interpretations) become *glued* together in the overall understanding of the building. *Guggenheim Bilbao* can be thus imagined as a sort of Grothendieck topos, both in metaphorical abstraction and in concrete particularization. Of course, mathematical exactness does not enter into the parallel (except for CATIA's implementations), but the *allegorical* impact is profound (for the force of allegories, *cfr.* [Benjamin 1927-1940]). In that line of thought, *Bilbao*'s *heart* –its non-rectifiable atrium– possesses exactly the same "transient quality" of the *heart* of a non-extensional Grothendieck topos, where its *subobject classifier* demonstrably carries plastic, non-classical logics. On another hand, the *reflection* of plasticity in a topos (an intermediate, intuitionistic, logic reflected *locally* in all the Heyting algebras of subobjects) may be viewed as *Bilbao*'s dance of foldings and unfoldings of the structure, along all the passages, rails, stairs, curves, which allow the aerial transit over the atrium.

2.3. Kiefer: residues of the unplumbed

Anselm Kiefer (b. 1945) has turned into the *Great Blacksmith* of the fleeting and contradictory times that we are living between millenniums. Since the end of XXth century to the beginning of XXIst, his forge attains the mythical dimensions of Vulcan and his work seems to perpetuate Prometheus' feats. Kiefer immerges himself completely in the domains of the *unplumbed*: realms of the imagination not previously sounded, measured, or explored with a plumb. His awesome *ateliers* in Buchen, Barjac, or Croissy [Cohn 2012], combine a multidimensional *conceptual* boldness and a polysemous plurality of *matter*, where thousands of fibers are folded and unfolded in violent alloys of the organic and the inorganic. Stone, earth, glass, lead, help to transform the many *residues* of our civilization in gigantic works of art (*Monumenta* 2007 is a good place to observe Kiefer's monumentality, *cfr.* [Kiefer 2007]). Grothendieck's stature is well mirrored in Kiefer's own achievements. Indeed, as difficult as it may be to signal prominent figures in art, undoubtedly Kiefer ranges among the ones that will survive our ruins (*cfr.* his lessons at the Collège de France, *L'art survivra à ses ruines* [Kiefer 2011]). Exploring the unplumbed, Grothendieck, Tarkovsky, Gehry, Kiefer, all lie among those that have changed our usual perspectives, and have helped to open our *blind eyes*.

For Kiefer, following Manichean and alchemical traditions, "the earth's sparks of light need to be extracted and elevated to the heavens until the earth, having no inner sun, perishes in darkness" [Kiefer 2011, pp. 298-299]. In fact, Kiefer's artistic processes may be understood as profound *transmutations* of the elements, looking for an elusive (and impossible) *pure abstract freeness*, after effecting heavy gestural immersions in *matter*. In the same way as Grothendieck, Tarkovsky or Gehry imagine aerial architectonics based on concrete materials, Kiefer is also an *architect* of the sublime rooted in a *muddy* reality, where *darkness* appears as the essential fountain of creativity (for similar ideas in Valéry, Warburg, and Florensky, see [Zalamea 2013]). After Auschwitz, the World can not be more coded in an illusory Book, but, through a *double transmutation* (not One, but Many – not positive, but negative), Kiefer tries to cypher the World along a troglodyte Library of lead, corroded books (see *Figure 4*). An architectonics is still possible, but only going back to mythical insights, archaic perspectives, or cosmic imagination (*The Seven Heavenly Palaces* [Kiefer 2001]). In that search, a complex web of *strata* is essential, in order to provide a sense of time and history for the work of art.

Figure 4
The Breaking of the Vessels (1990) [Kiefer 2007, p. 264]

The *multiplication of strata* (close to Warburg's ideas, *cfr.* [Zalamea 2013]) is also fundamental in a Grothendieck topos perspective. Layers, levels, sections, are in fact encoded in the very notion of a *sheaf* (recall *Figure 1*), but, even more interesting, they occur ubiquitously all along the *internal structure* of a topos: as witnesses of infinite towers in an exponential (cartesian closure of the topos), as approaching webs in limiting processes (limit closure of the topos), as logical traces in algebraic types (subobject classifier). In this sense, the *very axioms* of a topos (cartesian closure, existence of limits, subobject classifier) reflect a *continuous connection of layers*, which embodies Grothendieck's typical understanding of the main dialectics of mathematical thought (Many/One, discrete/continuous, algebraic/geometric, etc.) Some sort of *density* is necessary to capture the complexity of the world, as Kiefer also shows in his big canvases. *Orages de roses* (1998), *Ta Maison chevaucha la sombre vague* (2005), *Flocons noirs* (2005) (*cfr.* [Kiefer 2007, pp. 170-171, 206-207, 241]), introduce for example many techniques (shellac, emulsions, wires, flowers, lead, plaster, wood) which *expand* the fabric of the canvas towards *physical* three dimensions, and towards *spiritual* multiple dimensions. Kiefer's sites –closely woven, coarse, dense, thick, heavy– reflect thus *materially* and *metaphorically* a multiplication of strata, as happens *axiomatically* in a Grothendieck topos.

In an interview with Daniel Arasse (November 15 2001), Kiefer explains how matter and spirit are inextricably woven: "Let's talk about materiality, matter. For me, matter is related to the spiritual. I do not make a distinction between an idea and its execution. (...) I think that the spirit is in the things, in matter. In lead, in straw... One has to discover, to reveal the spirit that they harbor. It is a primordial question" [Kiefer, Arasse 2010, p. 65]. As we have seen, the same question is almost exactly stated by Tarkovsky, and it is also an essential question both for Grothendieck (interaction between mathematical free archetypes and concrete types) and for Gehry (interrelation between free space drawing and concrete implementation). *Freeness* is here the main concept at stake. *Creativity* needs both the most abstract and the most concrete. Our wanderings through mathematics, cinema, architecture, and art, have shown the importance of a *full* dialectics between materiality and non-materiality, through many perspective *inversions*. The pendulum has to go *back-and-forth* unremittingly, knowing, as Bakhtin well summoned, that all the important knowledge lies in the *frontiers*.

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